

NoisiQ: Noise-Aware Quantum Circuit Simulation and Visualization

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Abstract

Noisy intermediate-scale quantum processors are limited not only by aggregate error rates, but also by the physical structure, timing, and propagation behavior of the underlying noise channels. NoisiQ is an open-source Python package for noise-aware quantum circuit simulation and visualization that connects circuit-level algorithm design with hardware-inspired error mechanisms. The package provides an intermediate circuit representation, imports common circuit formats, and routes simulations according to circuit and noise structure: Clifford circuits with Pauli-compatible noise are handled through a stabilizer/Pauli-frame pathway built around Stim, while circuits requiring non-Clifford gates, Kraus channels, coherent rotations, or other non-Pauli effects are handled by an exact state or density-matrix simulation pathway. NoisiQ models Pauli noise, amplitude damping, dephasing, coherent over-rotations, correlated multi-qubit errors, readout and state-preparation errors, and hardware-profile-derived idle decoherence. Its visualization layer translates these mechanisms into interpretable diagnostics, including step-resolved error-propagation animations, gate-level heatmaps, idle-decoherence overlays, interactive circuit inspection panels, and exportable figures or animations. We present the architecture, supported noise models, diagnostic metrics, and representative demonstrations, including GHZ-state hardware-profile comparisons, dynamical decoupling under time-dependent decoherence, and mid-circuit-measurement workflows relevant to quantum error correction. NoisiQ is designed to support both research prototyping and pedagogy by making hardware-aware noise behavior visible at the circuit level.

1 Introduction

2 Design Philosophy

2.1 Modularity and Expansion

2.2 Teaching Tool

3 Software Architecture

3.1 Quantum Simulator

STIM

Qiskit

3.2 Noise Models

Hardware Noise

3.3 QASM

3.4 Visualization

Heatmap

Interactive Panel

Animation

4 NoisiQ Examples

4.1 Meier-Eastin-Knill (MEK) State Distillation

4.2 Greenberger-Horne-Zeilinger (GHZ) State

4.3 Dynamical Decoupling

Sequences

Hardware Noise Comparison

4.4 Quantum Fourier Transform

5 Metrics

6 Conclusion

Future Work

Acknowledgments

Supporting Examples / Code

Grover's Algorithm

Deutsch-Jozsa Algorithm

Quantum Phase Estimation